

CS 422 – CS303 Language Theory & Finite Automata

Project Assignment 2025-2026

Real-Life Application of Automata Theory

Project Title: Smart Home Security System Using Face Recognition

	Student Name	Student ID
Group Members:	1- Jana alshammary 2- Marwa Abdulaziz 3- Alya Shehab 4- Maha Aljoaib 5- Maisa Alzahrani	2240006642 2240000811 2240002062 2240006849 2240004660
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Instructor:	Leena Awad Alhusain	

2.1 Deterministic Finite Automaton (DFA)

2.1.1 DFA Components

$M = (Q, \Sigma, \delta, q_0, F)$:

$Q = \{q_0, q_1, q_2, q_3, q_4, \text{dead}\}$

$\Sigma = \{d, v, u, e\}$

$q_0 = q_0$ (initial state)

$F = \{q_4\}$ (Accepting/final state)

δ is defined as shown in the transition table below.

Alphabet key:

d = detect face

v = verify face

u = unlock door

e = enter home

2.1.2 Transition Table

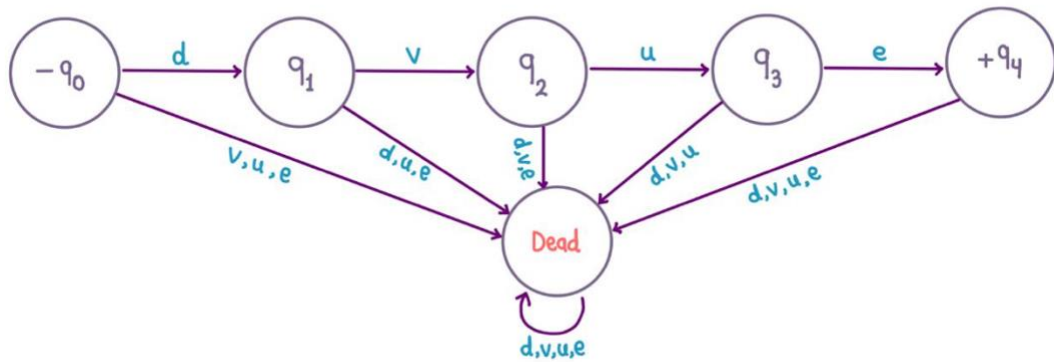
The transition function $\delta : Q \times \Sigma \rightarrow Q$ is defined as follows:

State	d	v	u	e
-q0	q1	dead	dead	dead
q1	dead	q2	dead	dead
q2	dead	dead	q3	dead
q3	dead	dead	dead	q4
+q4	dead	dead	dead	dead
dead	dead	dead	dead	dead

2.1.3 DFA Diagram

The DFA state diagram is shown below. The diagram must clearly indicate:

- Initial state q_0 (arrow pointing to it from outside)
- Accepting/final state q_4 (double circle)
- All transitions labeled with input symbols (d, v, u, e)
- Dead state q_{dead} with self-loops for all inputs



2.2 Grammar Production Rules

2.2.1 Grammar Definition

$$G = (V, \Sigma, R, S)$$

Where:

- $V = \{S, A, B, C\}$
- $\Sigma = \{d, v, u, e\}$
- $S = S$ (start symbol)
- R = set of production rules defined in Section 2.2.2

2.2.2 Production Rules

$$R = \{$$
$$S \rightarrow dA$$
$$A \rightarrow vB$$
$$B \rightarrow uC$$
$$C \rightarrow e$$
$$\}$$

2.2.4 Derivation Example

Step-by-step derivation of the valid string "d v u e":

Step	Derivation	Rule Applied
1	S	Start symbol
2	$S \Rightarrow dA$	Apply R1: $S \rightarrow dA$
3	$dA \Rightarrow dvB$	Apply R2: $A \rightarrow vB$
4	$dvB \Rightarrow dvuC$	Apply R3: $B \rightarrow uC$
5	$dvuC \Rightarrow dvue$	Apply R4: $C \rightarrow e$ (final string)

Complete derivation: $S \Rightarrow dA \Rightarrow dvB \Rightarrow dvuC \Rightarrow dvue$